T243 | Wk 11 | S07 | GP-S404 | Sarah

P631: Accelerating opposition analysis in professional rugby

Past fortnight

Over the past fortnight, the Team progressed from working mainly with assumptions and public rugby data to using the representative dataset and additional information provided by the Industry Partner. The prototype was updated to work directly with the Partner's event-data structure and now supports several opposition-analysis areas, including Kicking, Breakdown Choices, Tackling, Linebreaks, Game Intervals and Key Insights. The Team also improved the data-loading and processing workflow so that Partner XLSX/CSV files can be validated, transformed and analysed more consistently. Pasted text

Alongside the technical development, the Team reviewed the Partner's existing opposition preview and current workflow to better align the prototype with the analysis already used by the Partner. Technology research was updated using the actual Partner dataset, user stories and acceptance criteria were developed for key users, and additional project evidence was created through Tableau dataset exploration, a Kicking Analysis Dashboard, an updated Current Workflow Diagram, and a Prototype Gap Analysis. These activities helped the Team identify which Partner needs are already supported and which areas still require validation or further development. Pasted text

Fortnight ahead

Over the next fortnight, the Team will focus on validating and refining the current prototype rather than adding unnecessary features. We plan to test the prototype more closely with the Partner-provided dataset, confirm that the current analyses and outputs align with the Partner's workflow, and identify the most important remaining gaps before the final demonstration. Where additional Partner fixtures become available, we will also test whether the current processing pipeline can support multiple matches and begin exploring opponent-level trends and comparisons. Pasted text

The Team will also clarify remaining questions such as the Partner's field-zone definitions, preferred final output format, and whether the final solution should replace, complement, or export into the Partner's existing reporting workflow. User stories, documentation and supporting evidence will continue to be refined, while the Team prepares the Week 13 report and presentation and defines realistic, feasible and meaningful goals for Phase 2. Pasted text

To discuss

We would like feedback on whether the current prototype is addressing the most important Partner pain points and whether our current priorities are appropriate for the remainder of Phase 1. In particular, we would like to discuss which remaining gaps should be prioritised before the final demonstration, how much of the Partner's existing opposition preview should be reproduced or supported by the prototype, and what would be considered convincing evidence that our proposed Phase 2 goals are feasible and realistic.

We would also like guidance on whether the current level of Partner validation is sufficient, or whether we should prioritise obtaining further confirmation from the Partner on field-zone definitions, preferred output format, multi-match analysis and how the prototype should fit into the Partner's existing reporting process.

Partner Requirements and Prototype Gap Analysis

Partner Need / project consideration	Prototype coverage	Evidence / observation	Next step
Reduce manual data preparation	**Strongly supported**	The prototype automatically loads, validates and transforms Partner event data before running the analysis. This directly addresses the Partner's concern that putting the data together takes too much time.	Confirm which remaining preparation steps are still completed manually by the analysts.
Use Partner event data directly	**Supported**	The current prototype works directly with the representative Partner XLSX/CSV event structure rather than relying on the earlier public-data format.	Test the same pipeline with additional real Partner fixtures when available.
Reduce repeated manual analysis / notebook work	**Strongly supported**	Kicking, Linebreaks, Breakdown Choices, Tackling, Game Intervals and Key Insights are brought together in one interface instead of requiring separate analysis steps.	Confirm whether any important parts of the current analyst workflow still require separate notebooks or tools.

Kicking analysis	**Strongly supported**	The prototype provides kick types, outcomes, kickers, player positions, contest-status information and pitch locations.	Validate the Partner's exact definitions for metrics such as contestable, attacking and territorial kicks.
Breakdown Choices analysis	**Supported**	Playmaker Options are analysed by Pass, Carry and Kick together with receiver context and field location.	Confirm the official Reds field-zone definitions so the temporary coordinate bands can be replaced if required.
Tackling analysis	**Strongly supported**	The prototype includes tackle outcomes, missed tackles, player-level summaries, first-tackler analysis, dominant tackles, tackle height and tackle evasions.	Validate that the event definitions and player-level summaries match the Partner's current reporting approach.
Linebreak analysis	**Supported**	The prototype identifies recorded linebreaks achieved and conceded and provides player and location information where available.	Confirm whether the Partner needs additional linebreak measures such as phase or defensive-player information.
Visualise events on the rugby pitch	**Supported**	Recorded event coordinates are displayed on reusable pitch visualisations across several analysis modules.	Confirm whether the current pitch presentation is useful to analysts/coaches and whether additional visual interaction is needed.
Reduce time spent identifying useful information	**Partially supported**	Tables, summary metrics, charts and factual Key Insights are generated automatically from the processed events.	Ask the Partner which insights are most useful when preparing information for coaches and refine the summaries around those priorities.
Support multiple matches / opponent trends	**Not yet complete**	The system can distinguish fixtures and is designed to accept multiple files, but the current Partner data only provides one	Once additional Partner fixtures are available, add opponent-level aggregation, trends and match-to-match comparisons.

		representative fixture for validation.	
Automate the full opposition preview/report	**Partially supported**	The prototype now reproduces several major sections of the existing Partner preview interactively, but does not yet generate the full multi-match report.	Confirm which remaining preview sections are required and what the final output format should be.
Fit into the Partner's existing reporting workflow	**Still to be confirmed**	The prototype currently uses Streamlit, while the Partner's existing workflow uses an established analysis/reporting process.	Confirm whether the final solution should replace the current process, complement it, or export results into the Partner's existing reporting tools.

This section reviews the Partner’s known pain points and current workflow against the Team’s prototype to identify what is already supported, what remains incomplete, and what should be prioritised next.

a. Kelvin Dang (n12051632)

My work

Over the past fortnight, my main contribution has been continuing the technical development of the project after we received representative event data and additional information from the Queensland Reds. Earlier in the project, I used public rugby data mainly to understand match-event structure and build an initial proof-of-concept. Once the Partner data became available, I shifted my focus to working directly with the Partner event schema and aligning the prototype more closely with their existing opposition-analysis workflow.

I reviewed the representative dataset, Partner Q&A information and example opposition preview to understand what information was available and which parts of the current workflow could be reproduced reliably. Based on this, I created a separate Week 11 version of the prototype and reorganised the workflow into data loading, validation, transformation, analysis and visualisation so that the system can be reused with additional files of a similar structure.

I then developed and refined several analysis sections, including Kicking, Linebreaks, Breakdown Choices and Tackling. The prototype can now process the selected match and team, calculate relevant event statistics, show player-level information where available, and visualise event locations on a rugby pitch. I also added Game Intervals and Key Insights to provide a quicker overview of the recorded match data.

I improved the data-input workflow by allowing authorised XLSX or CSV files to be uploaded directly through Streamlit instead of relying on manual file paths or separate scripts. The prototype can identify fixtures and teams from the uploaded data, and the workflow is also prepared to handle multiple files separately as a foundation for future multi-match analysis.

I also developed the **Event Replay** feature, which allows users to move through recorded match events in sequence and view their locations on the field. I later added event-specific visual effects so that different event types are represented more clearly. Events with recorded start and end coordinates can show movement between those points, while tackles, missed tackles, breakdown events and other categories use different visual cues. This improves the demonstration and exploration of the data while still keeping the replay based on recorded events rather than continuous player tracking.

Another important part of my work was testing and documentation. I expanded the automated tests across the main data-processing, analysis and Event Replay functions, and manually checked important outputs against the representative Partner fixture. Following previous assessment feedback, I also improved the README and supporting documentation and organised the completed Week 11 prototype under my GitHub contribution folder so that the tutor can directly review the source code, tests and supporting evidence.

Overall, the prototype has progressed from an early public-data experiment to a more complete Partner-data workflow that can validate and transform event data, generate several opposition-analysis views and provide an interactive Event Replay. The remaining gaps mainly relate to additional Partner data, multi-match analysis and some Partner-specific definitions.

My plans

Over the next fortnight, I plan to focus more on validating the current prototype with the tutor and Partner rather than adding many new features. I want to demonstrate the workflow clearly and confirm whether the current analysis sections and visualisations are aligned with what the analysts actually need.

If more Partner match files become available, I plan to test whether the current pipeline can process them without match-specific changes and begin moving toward multi-match analysis, including opponent trends and summary statistics.

I also want to clarify several Partner-specific definitions that are still uncertain, particularly the official field zones and some analysis categories, before making further changes to the calculations.

For Event Replay, I plan to gather feedback on whether the current animations and event-specific visual effects are useful. I will also confirm the Partner's preferred final output format, including whether Streamlit should remain the main interface or whether a more traditional opposition preview/report should also be supported.

Finally, I will continue improving the documentation and GitHub evidence so that the current progress, remaining gaps and next steps are clear for both the tutor and the Partner.

My evidence: **Week 11 Partner-Data Analysis Prototype**

b. Leon Duong (n11849789)

My work: Technology_Research_and_Justification

Over the past fortnight, my work has mainly focused on moving from the assumptions we had in Version 1 to working with the actual data we have now received from the partner.

In Version 1, the partner had not provided the datasets yet, so most of my work was based on assumptions about what type of rugby data we might receive. I mainly focused on learning more about rugby and opposition analysis, thinking about what information coaches and analysts would need, and preparing questions for the partner. At that stage, it was difficult to make more specific technical decisions because we did not yet know the exact structure or fields in the data.

Now that we have received the new datasets, I have been able to look at the actual data structure and understand what information is available, including teams, players, events, match time, event types, results and field locations. I also compared the new partner data with Kelvin's existing prototype to see how the data could fit into his Python, Pandas, Streamlit and Plotly implementation.

From this, I worked on the Technology Research and Justification section, focusing on how we can connect the partner's dataset with Kelvin's existing code instead of rebuilding everything from scratch. I looked at the differences between the data structures and identified the need for a data normalisation step so that the partner data can be mapped into the format expected by the analysis code. I also looked at how the technology choices support the work of the other team members, especially the connection between the partner data, our analysis requirements and Kelvin's implementation. Compared with Version 1, I now have a much clearer understanding of what we can realistically build because we are working with the actual dataset rather than assumptions.

My plans

Over the next two weeks, my main focus will be on close coordination and discussion with the team to ensure everyone is aligned on the remaining tasks. I will prioritise completing and refining the outstanding work, particularly around the Technology Research and Justification section and its connection to the actual partner datasets and Kelvin's prototype.

At the same time, I will dedicate significant time to preparing for the group presentations. This includes organising the key points from our research, clarifying the technical

decisions we have made, and practising how to clearly communicate the progress and next steps to the audience. The goal is to finalise the content, support the team's overall presentation, and make sure our contributions are well-integrated and ready for delivery.

c. Nao Hakoda (n11917580)

My work

In the past fortnight I made user storis for an analyst, performance analyst, coach, and player. Also review the first version of the prototype.

- Analyst and performance analyst: I based these on the first partner meeting notes. Each story is written in a card format (role, biography, motivation, frustration, goals, user story, acceptance criteria, priority). The frustration links directly to the pain points I identified, such as the fragmented multi notebook process, repeated processing of already processed games, and manual review of many slides.
- Coach and players (indirect users): These users receive the outputs rather than operate the system. As they were not directly discussed in the meeting, I wrote about their frustrations and goals assumptions and marked them clearly.
- Acceptance criteria: I added acceptance criteria to each story, so that the team can check whether a feature or prototype meets the user's need.

Evidence: User Stories , Pain Points

My plans

In the next fortnight, I plan to update user stories based on feedback from the team, and on what we learn from the prototype and add some user stories as needed.

Also check that the user stories and acceptance criteria match the prototype directions the team is exploring.

I will also work on the report and presentation in week 13. Team will have a meeting about the content and discussed details, such as each member section and tasks.

d. Ngoc Ha Nguyen (n12248746)

My work

Over the past fortnight, I focused on possible prototype directions, Partner communication evidence, and data understanding and validation of the Partner's dataset.

To ground this work in the Partner's current practice, I reviewed the Partner's Fijian Drua opposition preview, which covers game intervals and ball-in-play, kicking, linebreaks, breakdown choices and tackling. The preview presents information at team, player and match-event levels. For example, the tackling section compares the opponent's defensive outcomes with competition-level results and identifies attacking players who frequently evade tackles or make offloads. This showed me that the prototype will need to support both overall comparisons and detailed player-level analysis.

For the prototype directions, I linked each idea to a pain point raised at our first Partner meeting on 4 September, so that our options are driven by the Partner's problems rather than by technology choices. I identified six possible small working prototypes, namely incremental extraction, combined extraction, one-click workflow orchestration, automated report generation, an automated data validation layer and an interactive opposition dashboard. I compared them on the pain point addressed, main user, fit with the current Partner data, build effort, demonstration value and dependency on the Partner. Because the Partner has so far provided one representative fixture, I recommended in my comparison that the Team integrate combined extraction, one-click orchestration and data validation into the Phase 1 demonstration, since these work fully with the data we have and build on the existing prototype. Incremental extraction and automated report generation are better suited to Phase 2, once more fixtures and the preferred output format are confirmed. These remain recommendations for Team discussion rather than a final technical decision.

For Partner communication evidence, I compiled a communication log covering our correspondence from our first email to the Partner through to the Partner providing the dataset. I also linked my summary of the first Partner meeting covering Dimitri's and Jack's workflows and pain points, documented the questions we prepared, and recorded the earlier follow-up email requesting the sample files and notebooks. In addition, I drafted a new follow-up email with clarification questions arising from the data validation.

For data understanding and validation, I reviewed the Reds versus Waratahs dataset, which contains 3,588 events across 88 columns for one fixture. I created a data dictionary

of the fields used by each analysis module and wrote a Python script that runs 29 repeatable checks. The event data accurately reproduces the final score of 26 to 17 and the half-time score, and the tackle and missed tackle records are consistent with each other. The checks also identified issues that affect our calculations. MatchTime is stored in minutes and seconds format, and all coordinates are relative to each team's attacking direction. In addition, 75.6% of end coordinates are recorded as (0,0), which appears to represent missing endpoint data and requires confirmation from the Partner, and 290 events contain action codes without corresponding descriptive labels. I also recalculated the figures in the Partner's Fijian Drua preview. Most sections reconcile fully, but I found three inconsistencies in the kicking section. I also identified definitions, such as how breakdown decisions are counted and the boundaries of the field zones, that need the Partner's confirmation before our outputs can be compared with theirs.

My evidence: Partner Meeting 1 Summary, Prototype Directions and Partner Communication, Data Understanding and Validation, Validation Script, Validation Results

My plans

Over the next fortnight, I plan to discuss the prototype comparison with the Team and help agree on which working component we will demonstrate at the end of Phase 1. Once the Team has reviewed the follow-up email, I will send it to the Partner and update our data dictionary and calculations based on their answers, particularly the field zones, the breakdown counting rule and the meaning of (0,0) end coordinates.

I also plan to work with Kelvin to build the validation checks into the prototype's upload workflow, so that every Partner file is checked automatically before analysis. If additional fixtures become available, I will rerun the validation on each file and confirm that our metrics match the Partner's preview. Finally, I will keep the communication log and evidence up to date on GitHub and contribute the prototype comparison and validation findings to the Week 13 report and presentation

e. Star Kamonthitiwuth (n11864133)

My work

Over the past fortnight, I continued contributing to the Team's project analysis, documentation, and prototype planning after the Industry Partner provided a representative event dataset.

Once the Partner dataset became available, I reviewed the information provided in the shared Q&A document and explored the structure of the representative Reds vs Waratahs event file. This allowed me to compare our earlier understanding of the Partner's workflow with the actual event-level data provided for the project.

I first conducted a Partner Dataset Exploration in Tableau to understand what information was available and how it could support opposition analysis. I created three visualisations: Event Count by Action, Team Event Comparison, and Event Locations. I then combined these into an interactive dashboard titled Partner Dataset Exploration – Reds vs Waratahs. The dashboard supports exploration by team and selected action types and helped me examine event frequency, differences between Queensland Reds and NSW Waratahs, and the spatial information available through the x/y coordinate fields.

Evidence: Partner Dataset Exploration – Tableau Dashboard

After completing the initial dataset exploration, I reviewed the Partner's Fijian Drua opposition preview together with the Industry Partner meeting information to understand how the Partner currently presents and uses opposition-analysis outputs. I identified that the existing preview includes analysis areas such as kicking, linebreaks, breakdown choices, tackling, and game intervals, with particular use of player, outcome, position, and spatial information.

Based on this review, I extended my Tableau work by creating a Kicking Analysis Dashboard aligned more closely with the Partner's existing opposition-analysis approach. I developed separate views for Kick Type, Kick Outcome, Kick Locations, Top Kickers, and Kicks by Position using fields from the representative Partner dataset. I also added a team filter so that the analysis can be viewed for Queensland Reds or NSW Waratahs. For the Kick Locations view, I mapped the Partner's x/y coordinates onto a rugby field background to make the spatial information easier to interpret.

Evidence: Kicking Analysis Dashboard

This second dashboard represents a progression from general dataset exploration toward a more focused opposition-analysis view. Rather than attempting to reproduce every page of the Partner's existing preview, I selected analyses that could be supported by the fields available in the representative dataset. This helped demonstrate how Partner data could potentially be presented in a more interactive and reusable format while remaining aligned with the current analysis process.

I also developed a Prototype Gap Analysis / Project Status Review for the Team. I compared the known Partner needs, workflow pain points, and project considerations against the current prototype to identify which areas are already supported, partially supported, still in progress, or not yet addressed. I included proposed next steps so that the Team could more clearly identify what should be prioritised as the prototype develops, and I asked Team members working directly on implementation to update any statuses that had changed.

After reviewing the Partner dataset, I also updated my Current Workflow Diagram to Version 2. The original diagram was mainly reconstructed from the first Industry Partner meeting, while version 2 uses the representative dataset to refine and partially validate the event data component of the workflow. I updated the input from general game CSV files to structured event CSV files and documented that the data contains event-level information including team/player details, timestamps, spatial coordinates, actions, results, and qualifiers. I retained the previously identified workflow pain points while making a clearer distinction between Dimitri's data-analysis workflow and Jack's performance and footage-analysis workflow.

Evidence: Current Workflow Diagram v2

Overall, these activities helped me move from understanding the Partner's current process, to exploring the actual Partner data, and then to investigating how that data could support a more focused opposition-analysis interface. They also helped me identify how the Team's prototype can be evaluated against the Partner's current workflow and pain points.

My plans

Over the next fortnight, I plan to use the updated workflow, Prototype Gap Analysis, Partner dataset exploration, and Kicking Analysis Dashboard to support the Team in reviewing the direction of the final prototype.

I plan to test the Team's updated prototype more closely with the Partner-provided dataset and check whether the current data-processing and visualisation functions work reliably with the Partner's actual event structure. I will also compare the prototype outputs against the analysis areas identified in the Partner's existing opposition preview to help identify any remaining gaps that should be prioritised.

I also plan to continue refining the Tableau analysis where it provides useful evidence for the project, particularly where it can help clarify the information that analysts may need to view, filter, or compare. However, I will avoid adding visualisations only for presentation purposes and will focus on analyses that can be linked to the Partner's workflow, data, or identified needs.

As the Team moves closer to the final demonstration and Phase 2 planning, I will continue maintaining my project evidence in GitHub, update the workflow or gap analysis where necessary, and support the Team in defining realistic remaining goals. I will also help ensure that the final solution can be clearly explained in terms of how it reduces or simplifies parts of the Partner's current opposition-analysis process.